



Lab01

REPORT

Applied Machine Learning - COS30082

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A. Data Loading and Initial Exploration

- Firstly, I download dataset named *Titanic_Dataset.csv* from the following Kaggle link (<https://www.kaggle.com/datasets/yasserh/titanic-dataset>). Next, I display all data in the file.

```
## Import Pandas and Numpy Library. ##
import numpy as np
import pandas as pd
## Import .csv File. ##
titanic = pd.read_csv('Titanic_Dataset.csv')
## Print the Dataset. ##
titanic
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikinen, Miss. Laina	female	26.0	0	0	STON/O2 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
...	...	...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W/C. 6607	23.4500	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.7500	NaN	Q

891 rows x 12 columns

Figure 1: Titanic dataset preview loaded in Python

- Then, I used the *.info()* method to display the column names, non-null value counts, and data types of the dataset.

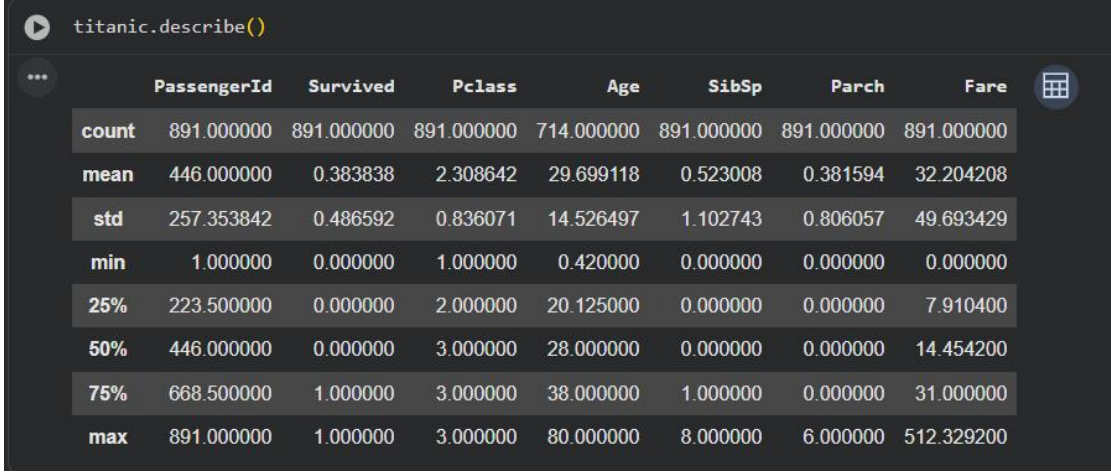
```
titanic.info()
```

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	object
4	Sex	891 non-null	object
5	Age	714 non-null	float64
6	SibSp	891 non-null	int64
7	Parch	891 non-null	int64
8	Ticket	891 non-null	object
9	Fare	891 non-null	float64
10	Cabin	204 non-null	object
11	Embarked	889 non-null	object

dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

Figure 2: Titanic dataset info and data types preview

- By using the `.info()` method, I see that there are 12 columns or features distinctively including PassengerID, Name, Pclass, and so on. Besides that, the method also highlights how many non-null values in each feature and data type which helps identify whether any missing values exist. As well as the data type of column. For instance, the column PassengerId has 891 values non-null value and has Integer type. Consequently, I utilized `.describe()` for calculating and showing descriptive statistics for the numerical columns.



	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

Figure 3: Descriptive statistics of the Titanic datasets

- In Drawing Graph part, I decide to draw a bar chart to showcase relationship between Sex(Gender => X) and Survived(Y). Based on the bar chart, it showcases that regarding female gender, more than 200 people survived while near ly 100 were not. In constrast, the ratio of not survived of male sex significantly reacher from 450 to 460 people while survived gained precisely at 100 people.



Figure 4: Survival count by Gender Distribution

Dataset and Visualization Summary:

The Titanic dataset highlights information on 891 passengers with the following features of each customer including passenger class, gender, age, and survival status. Moreover, the bar chart above showcases the relationship between gender and survival outcomes, clearly revealing a significant disparity where female passengers had a substantially higher survival ratio compared to male passengers.

B. Extended Data Exploration and Analysis

First and foremost, I generate pie charts to showcase the relationship between *Pclass* and *Survived* among three classes including First, Second, Third.

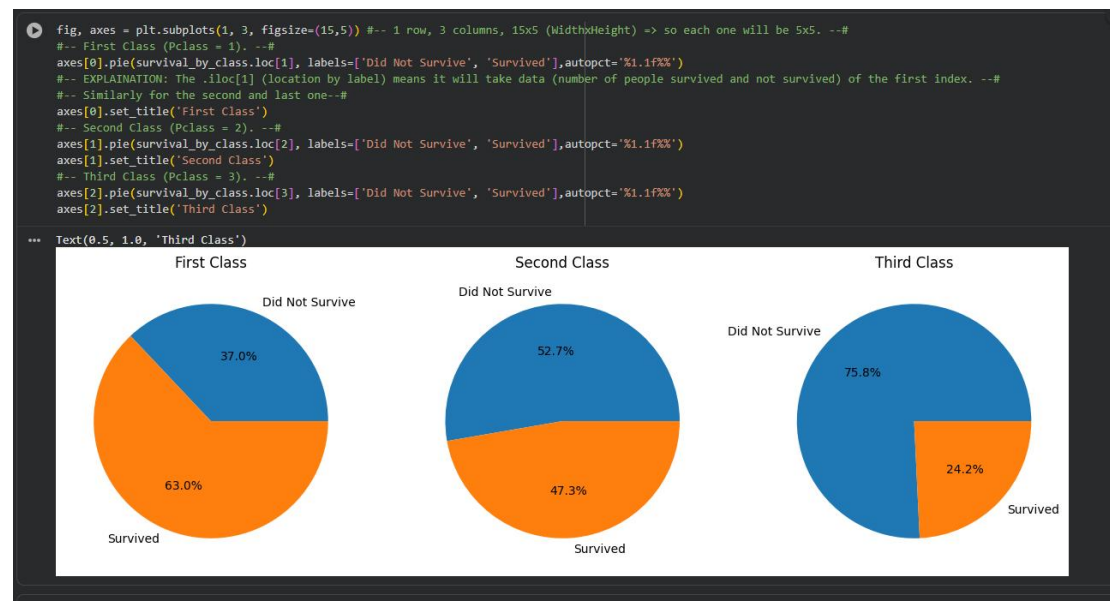


Figure 5: Survival Distribution by Passenger Class

The above pie charts show that the percentage of the first class for survived was 63% while not survived less than 26%. Furthermore, the number of did not survive and survived belongs to second class were 52.7% and 47.3% respectively. Lastly, the survived ratio of third class took one-fourth of total, particularly 24.2%.

Additionally, I use the histogram to highlight which age group has most majority in the dataset.

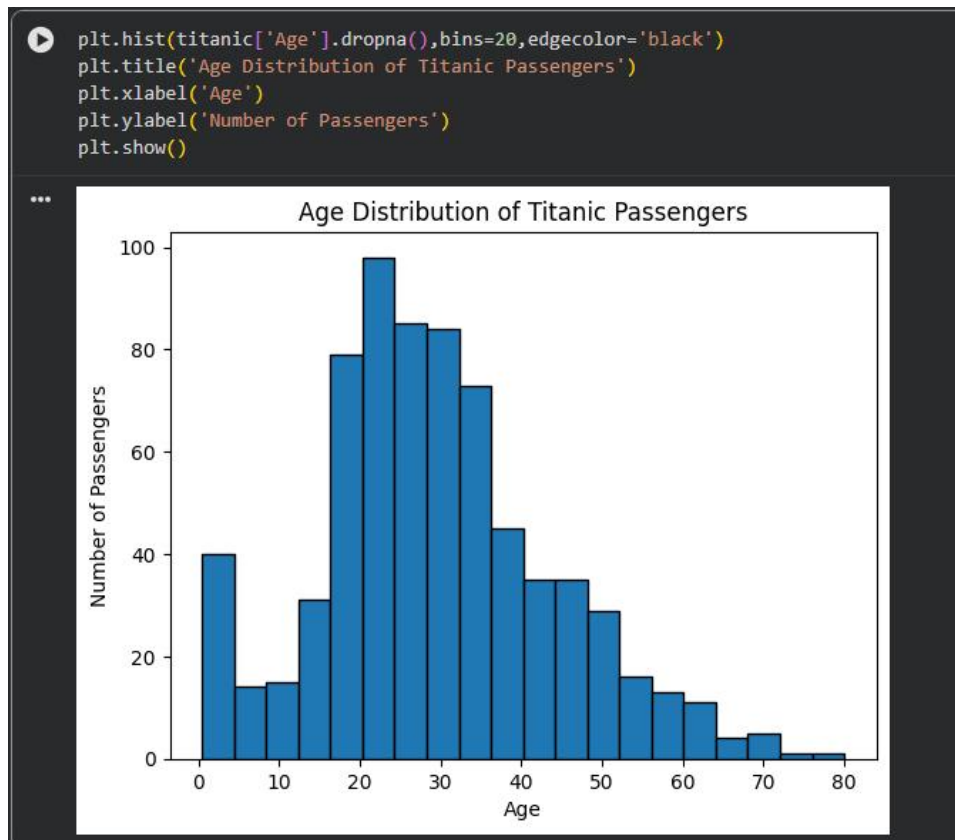


Figure 6: Age Distribution of Titanic Passenger.

In addition, I use the boxplot to check the Age effect or not to the survival rate. It is clear that age range from 20 to between 30 and 40 has most majority of the dataset

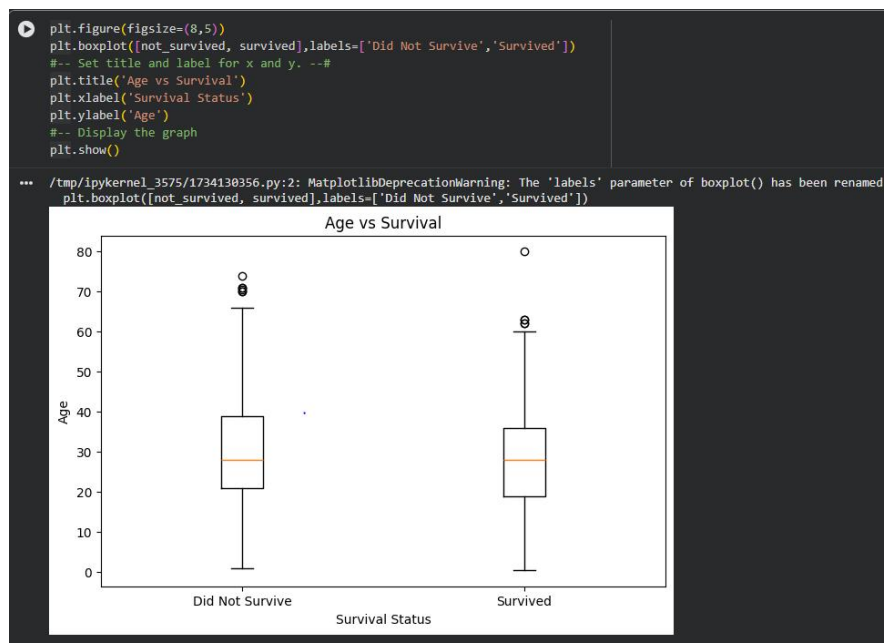


Figure 7: Age Distribution by Survival Status

In conclusion, after completing the first lab of Week 1 and conducting additional exploratory data analysis, I gained valuable insights into the dataset. Moreover, this hands-on practical lab allowed me to become familiar with several NumPy and Matplotlib functions and develop my understanding of data analysis at the data insight level such as how to draw bar graph basically. Furthermore, the additional exploratory data analysis provided valuable knowledge on how to explore data, interpret visualisations, and extract meaningful insights from datasets. In particular, I checked two factors including Passenger Class and Age how effect to Survival factor and Age Distribution through dataset by using boxplot.